Linear Algebra

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1 Vector Spaces

1.1 Operations

Definition 1.1.1 (Vector Space)

A group $(+, 0, -)$ and a linear operator (with the 0) $\cdot : \mathbb{F} \times V \to V$ satisfy:

- $\alpha(\beta v) = (\alpha\beta)v$
- $1v = v$, $0v = 0$, $\alpha 0 = 0$, $(-1)v = -v$
- $\alpha(v + w) = \alpha v + \alpha w$, $(\alpha + \beta)v = \alpha v + \beta v$

Definition 1.1.2 (Subspace)

$U \subseteq V$ is a subspace if $0 \in U$ and U is closed under $+$ and $\cdot$.

Proposition 1.1.3

Given two subspaces U, W of V, $U \cup W$ is a subspace if $U \subseteq W$ or $W \subseteq U$.

Proposition 1.1.4

Given two subspaces U, W of V, $U + W$ is the smallest subspace containing U and W.

Proposition 1.1.5

Given two subspaces U, W of V, $U \cap W$ is a subspace.

Definition 1.1.6 (Direct Sum)

V is the direct sum of U, W, $V = U \oplus W$, if $U \cap W = \{0\}$ and $U + W = V$.

Proposition 1.1.7

$V = U \oplus W$ if and only if every $v \in V$ can be decomposed into $v = u + w$ for $u \in U$, $w \in W$.

Proposition 1.1.8

$V = U \oplus W$ if and only if $\dim(V) = \dim(U) + \dim(W)$ and $V = U + W$.

Proposition 1.1.9

$V = U \oplus W$ if and only if $\dim(V) = \dim(U) + \dim(W)$ and $U \cap W = \{0\}$.

Proposition 1.1.10

Given U, W subspaces of V, $\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W)$.

Definition 1.1.11 (Span)

$\operatorname{span}(v_1, \ldots, v_n) = \{\alpha_1 v_1 + \cdots + \alpha_n v_n \mid \alpha_i \in \mathbb{F}\}$.

Proposition 1.1.12

$\operatorname{span}(v_1, \ldots, v_n)$ is the smallest subspace containing $v_1, \ldots, v_n$.

Definition 1.1.13 (Linear Independence)

$v_1, \ldots, v_n$ are linearly independent if $\alpha_1 v_1 + \cdots + \alpha_n v_n = 0$ implies $\alpha_1 = \cdots = \alpha_n = 0$.

Proposition 1.1.14

$v_1, \ldots, v_n$ are linearly independent if and only if each vector is not in the span of the previous vectors.

Lemma 1.1.15 (Linear Dependence Lemma)

If $v_1, \ldots, v_n$ are linearly dependent, then $v_k \in \operatorname{span}(v_1, \ldots, v_{k-1})$ for some k.

Proposition 1.1.16

$v_1, \ldots, v_n$ are linearly independent if and only if each v_k has a unique representation in $\operatorname{span}(v_1, \ldots, v_n)$.

Definition 1.1.17 (Basis)

$v_1, \ldots, v_n$ form a basis of V if $\operatorname{span}(v_1, \ldots, v_n) = V$.

Proposition 1.1.18

$v_1, \ldots, v_n$ form a basis of V if and only if every $v \in V$ can be uniquely represented as $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$.

Proposition 1.1.19 (Basis Reduction)

$v_1, \ldots, v_n$ span V. Some subsequence forms a basis for the subspace V.

Proposition 1.1.20 (Independent Complement)

$U \subseteq V$. W is an independent complement of U if $U \oplus W = V$.

1.2 Multiplication and Representation

Definition 1.2.1 (Linear Map)

$T : V \to W$ is a linear map if $T(v + w) = Tv + Tw$ and $T(\alpha v) = \alpha Tv$.

The operations are found from the characteristic equations.

Proposition 1.2.2 (Linear Map Lemma)

Given $v_1, \ldots, v_n$ basis of V and $w_1, \ldots, w_n$ in W, there exists a linear map $T : V \to W$ such that $Tv_i = w_i$.

Definition 1.2.3 (Null Space)

$\operatorname{null}(T) = \{v \in V \mid Tv = 0\}$.

$\operatorname{null}(T)$ is a subspace of V.

T is injective if and only if $\operatorname{null}(T) = \{0\}$.

Definition 1.2.4 (Range)

$\operatorname{range}(T) = \{Tv \mid v \in V\}$.

$\operatorname{range}(T)$ is a subspace of W.

T is surjective if $\operatorname{range}(T) = W$.

Theorem 1.2.5

$\dim(\operatorname{null}(T)) + \dim(\operatorname{range}(T)) = \dim(V)$.

Proposition 1.2.6

$T : V \to W$ is invertible, iff $\operatorname{null}(T) = \{0\}$ and $\operatorname{range}(T) = W$.

Proposition 1.2.7 (Properties)

- $\operatorname{null}(ST) \supseteq \operatorname{null}(T)$
- $\operatorname{range}(ST) \subseteq \operatorname{range}(S)$
- $\operatorname{null}(ST) = \operatorname{null}(T)$ if S is injective
- $\operatorname{range}(ST) = \operatorname{range}(S)$ if T is surjective

Proposition 1.2.8

T is invertible if and only if T is injective and surjective.

Proposition 1.2.9 (Isomorphic)

V, W are isomorphic if there exists $T : V \to W$ invertible.

Proposition 1.2.10

V, W are isomorphic if and only if $\dim(V) = \dim(W)$.

Proposition 1.2.11 (Isomorphic Vector Spaces)

Every finite dimensional vector space is isomorphic to $\mathbb{F}^n$.

Proposition 1.2.12

$\dim(\operatorname{null}(ST)) \leq \dim(\operatorname{null}(S)) + \dim(\operatorname{null}(T))$.

Proposition 1.2.13

$\dim(\operatorname{range}(ST)) \leq \dim(\operatorname{range}(S)) + \dim(\operatorname{range}(T))$.

Proposition 1.2.14

$\operatorname{null}(T^k) = \operatorname{null}(T^{k+1})$ implies $\operatorname{null}(T^k) = \operatorname{null}(T^{k+1}) = \operatorname{null}(T^{k+2})$.

Proposition 1.2.15

$\operatorname{range}(T^k) = \operatorname{range}(T^{k+1})$ implies $\operatorname{range}(T^k) = \operatorname{range}(T^{k+1}) = \operatorname{range}(T^{k+2})$.

Proposition 1.2.16

$\operatorname{null}(T^k) = \operatorname{null}(T^{k+1})$ implies $\operatorname{range}(T^k) = \operatorname{range}(T^{k+1})$.

Proposition 1.2.17

$\operatorname{range}(T^k) = \operatorname{range}(T^{k+1})$ implies $\operatorname{null}(T^k) = \operatorname{null}(T^{k+1})$.

Proposition 1.2.18

$\operatorname{null}(T^k) \cap \operatorname{range}(T^k) = \{0\}$ implies $\operatorname{null}(T^k) = \operatorname{null}(T^{k+1})$ and $\operatorname{range}(T^k) = \operatorname{range}(T^{k+1})$.

Proposition 1.2.19

$\operatorname{null}(T^k) + \operatorname{range}(T^k) = V$ implies $\operatorname{null}(T^k) = \operatorname{null}(T^{k+1})$ and $\operatorname{range}(T^k) = \operatorname{range}(T^{k+1})$.

Proposition 1.2.20

$\operatorname{null}(T^k) = \operatorname{null}(T^{k+1})$ and $\operatorname{range}(T^k) = \operatorname{range}(T^{k+1})$ implies $\operatorname{null}(T^k) \cap \operatorname{range}(T^k) = \{0\}$ and $\operatorname{null}(T^k) + \operatorname{range}(T^k) = V$.

1.3 Norms

Definition 1.3.1 (Norm)

$\|\cdot\| : V \to \mathbb{R}$ satisfies:

- $\|v\| \geq 0$
- $\|\alpha v\| = |\alpha| \|v\|$
- $\|v + w\| \leq \|v\| + \|w\|$

Proposition 1.3.2

$\|v - w\| \geq \|v\| - \|